

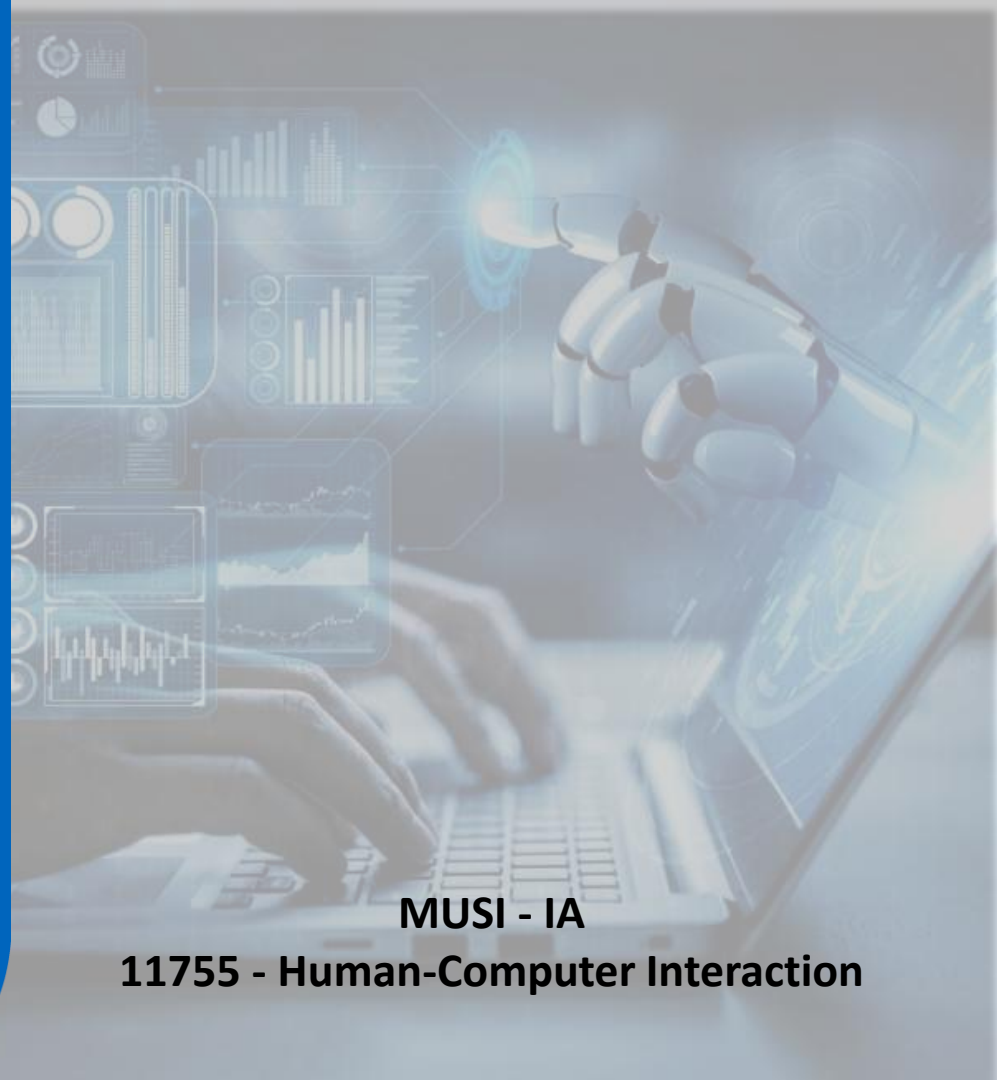


Universitat
de les Illes Balears

02 - Cognitive Psychology in HCI

Alejandro Mesejo

3-3-2025



MUSI - IA

11755 - Human-Computer Interaction

Cognitive Psychology, why?

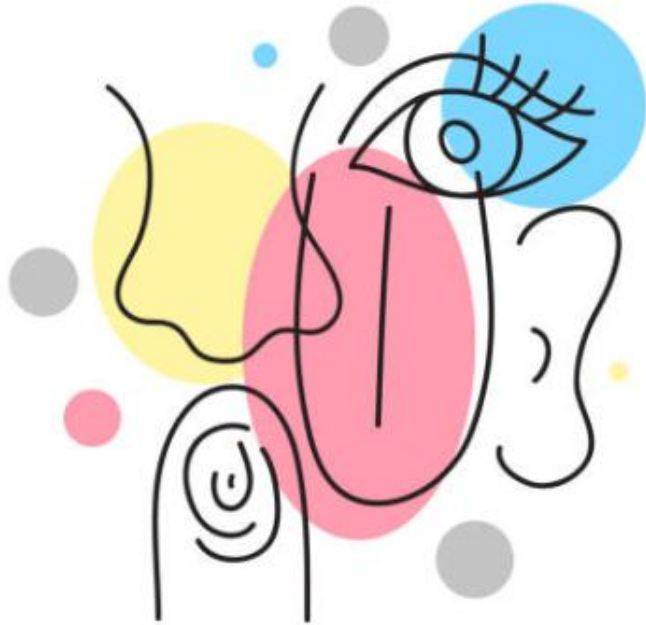
- In order to design something for someone, we need to **understand their capabilities** and limitations.
- We need to know if there are **things that they will find difficult** or, even, impossible.
- It will also help us to **know what people find easy** and how we can help them by encouraging these things.
- We will look at aspects of **cognitive psychology** which have a bearing on the use of computer systems: **how humans perceive the world around them, how they store and process information and solve problems, and how they physically manipulate objects.**



Universitat
de les Illes Balears

1

**How human
perception and
attention affect
interface use?**



Perception

- Perception **interprets** sensory information to form a mental representation of the world. It's influenced by experience, expectations, attention and varies across individuals.
- Sensory organs are the gateway to perception. They receive information from the environment and convert it into electrical signals processed by the brain (**sensation**).
- Each sense has its own specialized organ: the eyes for vision, the ears for hearing, the skin for touch, the tongue for taste, and the nose for smell.

Sensation

- Sensation refers to the ways that information from the world is picked up by sense organs and detected by the owners of those organs.
- Perception and sensation are viewed as different because sensation is seen as more “physical,” while perception is believed to be absolutely psychological.
 - perception is the process by which the brain selects, organizes, and interprets sensations.
- Sensory receptors are specialized neurons that respond to specific types of stimuli. When sensory information is detected by a sensory receptor, sensation has occurred.
 - For example, light that enters the eye causes chemical changes in cells that line the back of the eye. These cells relay messages, in the form of action potentials to the central nervous system.

Sensation, absolute threshold

- The sensitivity of a given sensory system to the relevant stimuli can be expressed as an **absolute threshold**.
- Absolute threshold refers to the **minimum amount of stimulus energy that must be present for the stimulus to be detected 50% of the time**.
 - Another way to think about this is by asking how dim can a light be or how soft can a sound be and still be detected half of the time.
- The sensitivity of our sensory receptors can be quite amazing. The farthest from which an average unaided human could see a candle is [about 1.6 miles](#).
- Absolute thresholds are generally measured under incredibly controlled conditions in situations that are optimal for sensitivity.

Sensation, Just Noticeable Difference (JND)

- The JND (or difference threshold) is the **minimum amount by which stimulus intensity must be changed in order to produce a noticeable variation in sensory experience.**
- Ernst Weber (in the 1830s) observed that the size of the JND appeared to be lawfully related to initial stimulus magnitude. This relationship, known since as Weber's Law, can be expressed as:

$$\frac{\Delta I}{I} = \frac{\text{JND}}{I} = k$$

where I represents the initial stimulus intensity and k is a constant.

- Weber's Law means that the stronger the initial stimulus is, the harder it is to notice small deviations from it.

Attention

- Is the ability to actively process specific information in the environment while tuning out other details.
- Let's us focus on what's relevant and tune out what's not. We use attention to learn more about our environment, form memories, and perform tasks.
- Is limited in terms of both capacity and duration. In other words, we can only focus on so much for so long.
- Not all types of attention are the same:
 - Sustained attention allows us to focus for long periods while alternating attention allows us to switch between tasks.
 - Selective attention allows us to choose what we want to attend to
 - Focused attention helps us notice stimuli that require an immediate response.

Effect of culture on perception and attention

- Our cultural background can influence how we interpret and organize sensory information.
- For example, color meaning differ across cultures, significantly affecting how we communicate and interact with people from different backgrounds

Interaction Design Foundation - IxDF. [“What is Perception in UX/UI Design?”](#)

Meaning of Colors in Cultures

	Western/ American	Japanese	Hindu	Native American	Chinese	Asian	Eastern European	Muslim	African	South American
Anger	Red		Black				Red		Red	
Art / Creativity	Black		Blue							
Authority	Black									
Bad Luck		Orange		Black		Green				
Balance		Orange								
Beauty	Purple									
Calm										
Celebration	Pink				Black					
Children	Pink					White				
Cold	Pink			Blue						
Compassion	Pink		Green							
Courage	Red		Orange				Red			
Cowardice	Yellow									
Cruelty	Red									
Danger	Red			Yellow						
Death	Black			Black	White			Blue		Green
Decadence	Purple									
Deceit	Orange									
Desire	Orange									
Earth	Brown				Brown					
Energy		Red	Red							

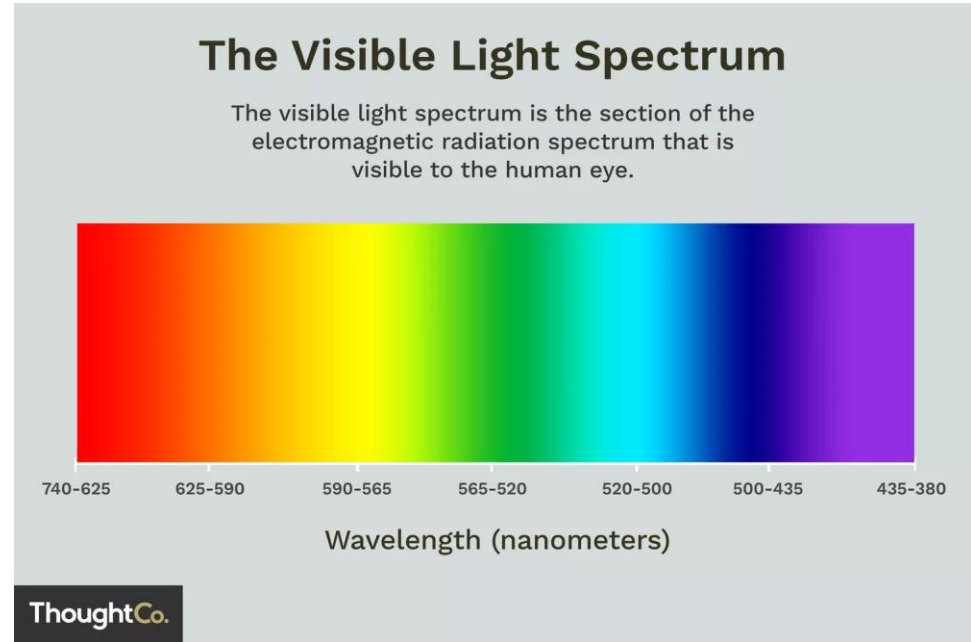
[illegible][illegible]

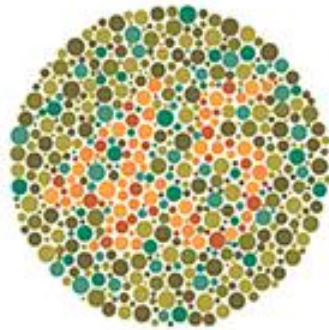
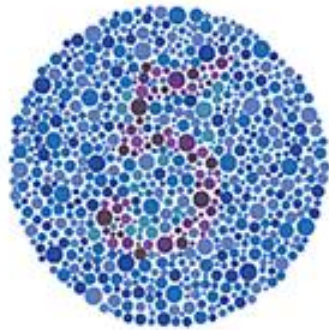
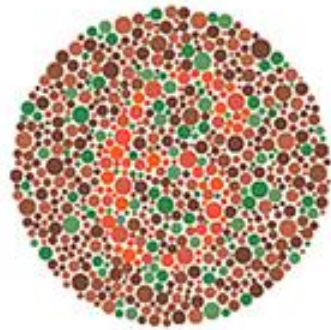
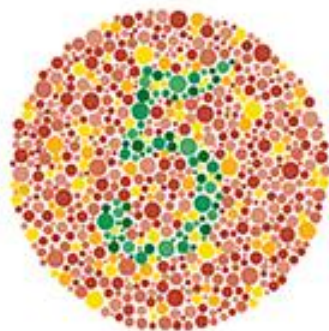
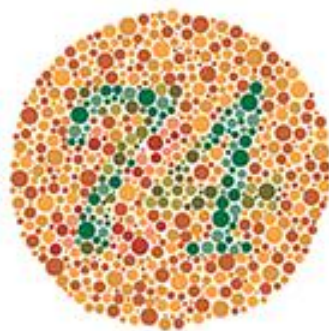
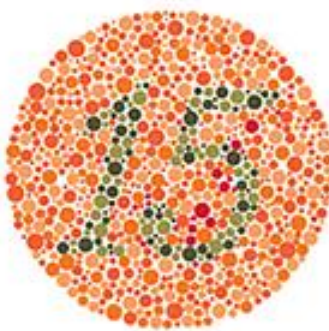
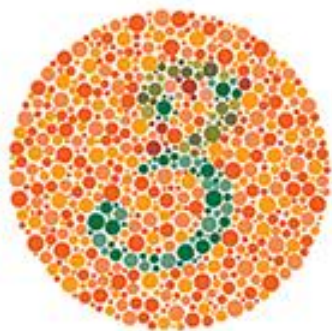
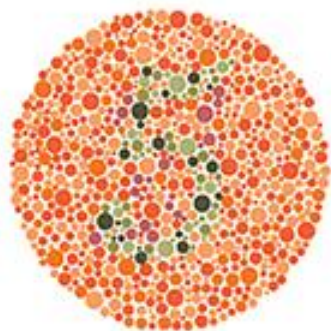
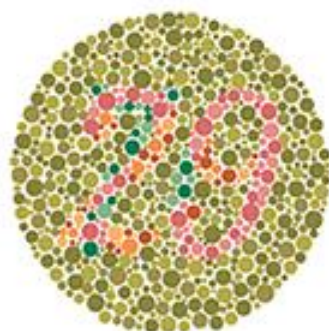
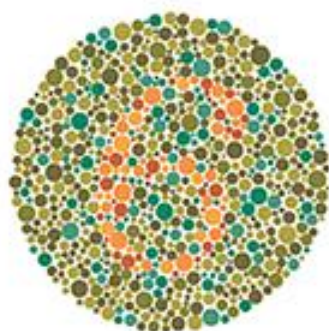
	Western American	Japanese	Hindu	Native American	Chinese	Asian	Eastern European	Muslim	African	South America
Penance			Black							
Power						Black				
Personal Power	Purple		Yellow			Yellow	Red			
Purity								Blue		
Radicalism	Red						Red			
Rational		Blue								
Reliable	Brown									
Repels Evil	Yellow	Red	Yellow		Green	Yellow				
Respect	Gray			Blue						
Royalty					Yellow					
Self-cultivation				Black		Black				
Strength		Red				Yellow		Yellow		
Style	Black	Black								
Success				Red	Red	Red	Red		Green	Red
Trouble				Blue						Blue
Truce					White					
Trust										
Unhappiness	Blue	Black		Blue						
Virtue	Purple									
Warmth		Orange								
Wisdom			Purple	Brown			Blue			

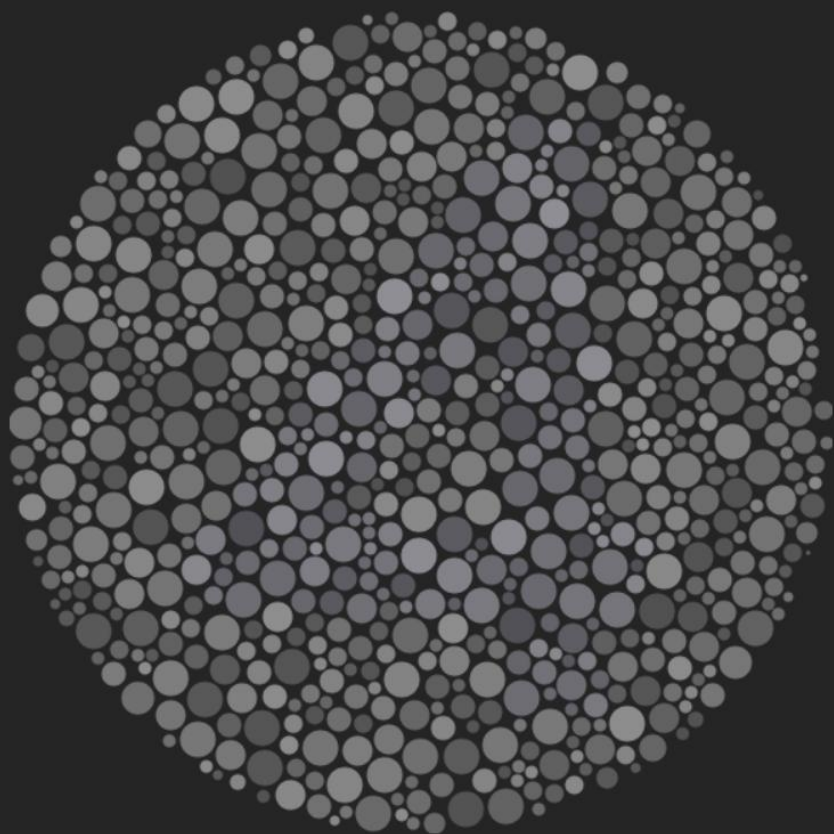
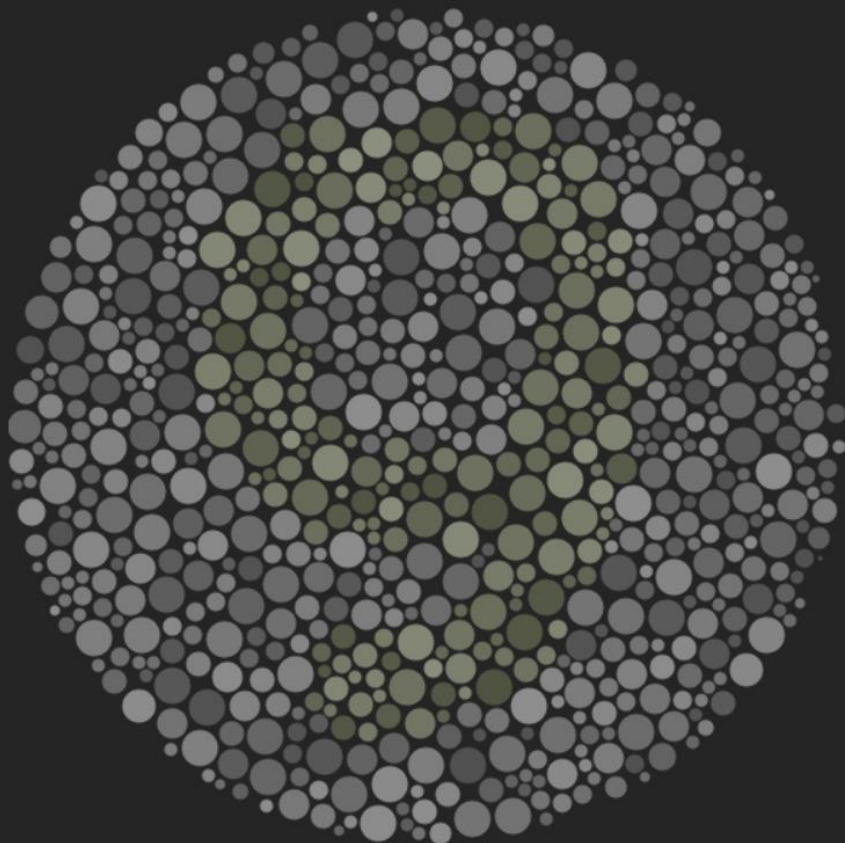
Sensation, Perception and Attention are limited

Sensory limitations:

- We can only detect a narrow band of electromagnetic radiation (visible light)
- Our hearing range is limited (roughly 20Hz-20kHz)
- We can't detect many environmental stimuli that other animals can
- Color blindness affects approximately 300 million people worldwide.



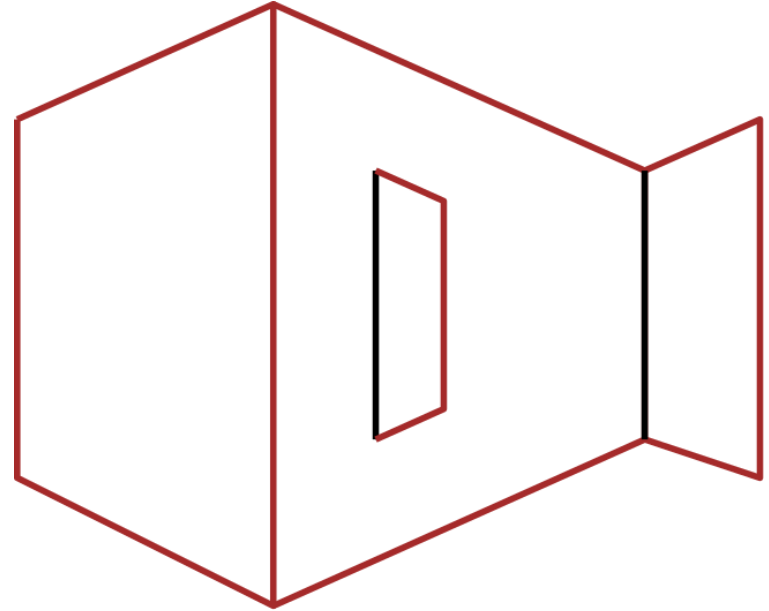




Sensation, Perception and Attention are limited

Perceptual limitations:

- Change blindness: we often fail to notice significant visual changes
- Inattention blindness: we miss obvious events when focusing elsewhere
- Our brains constantly fill in gaps and make assumptions about what we're perceiving
- Optical illusions demonstrate how our perceptual systems can be systematically fooled

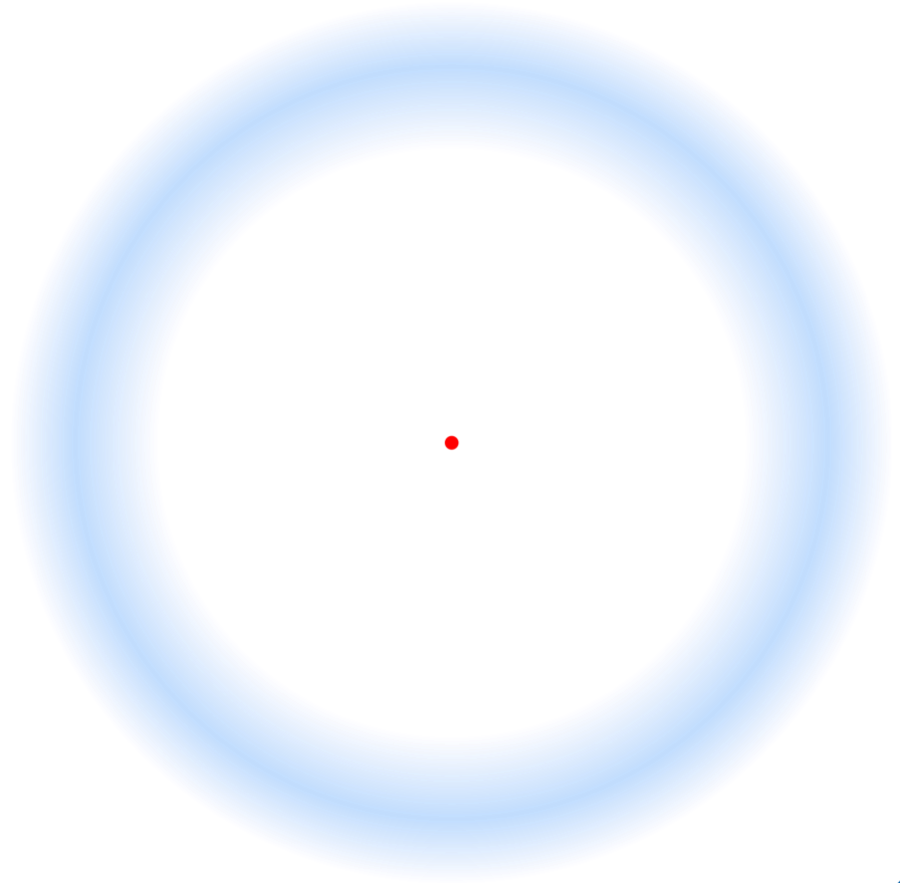


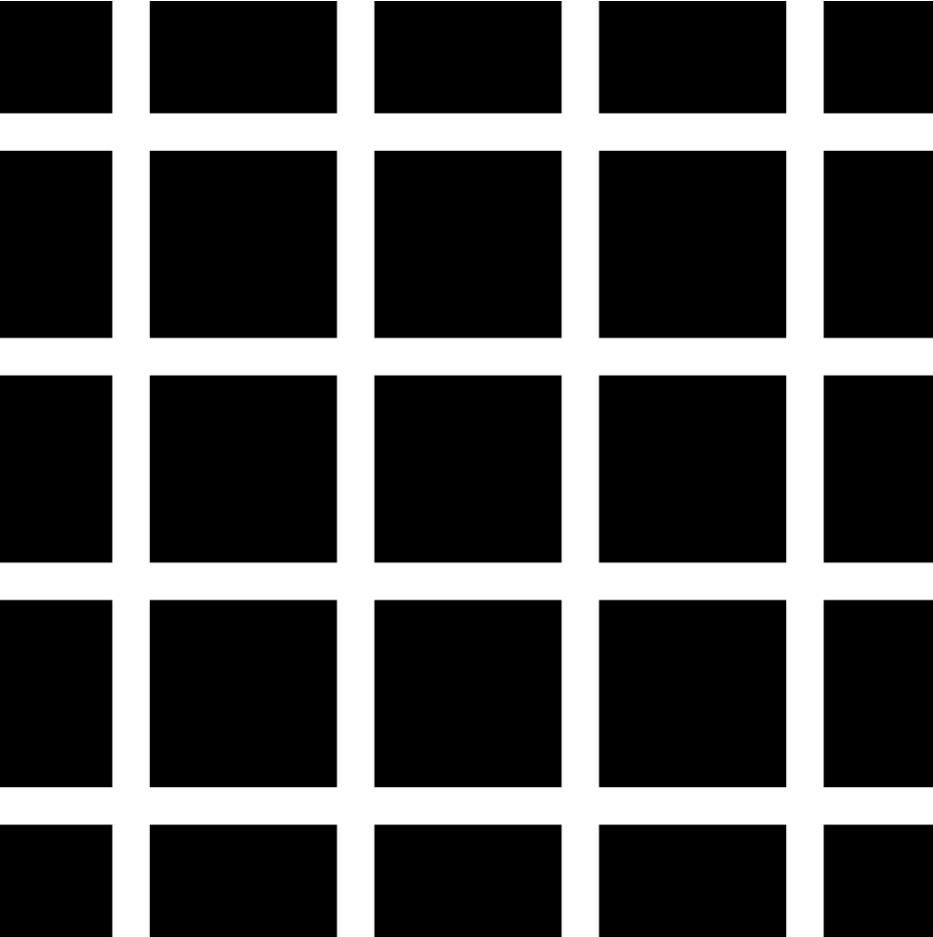
Müller-Lyer optical illusion

Sensation, Perception and Attention are limited

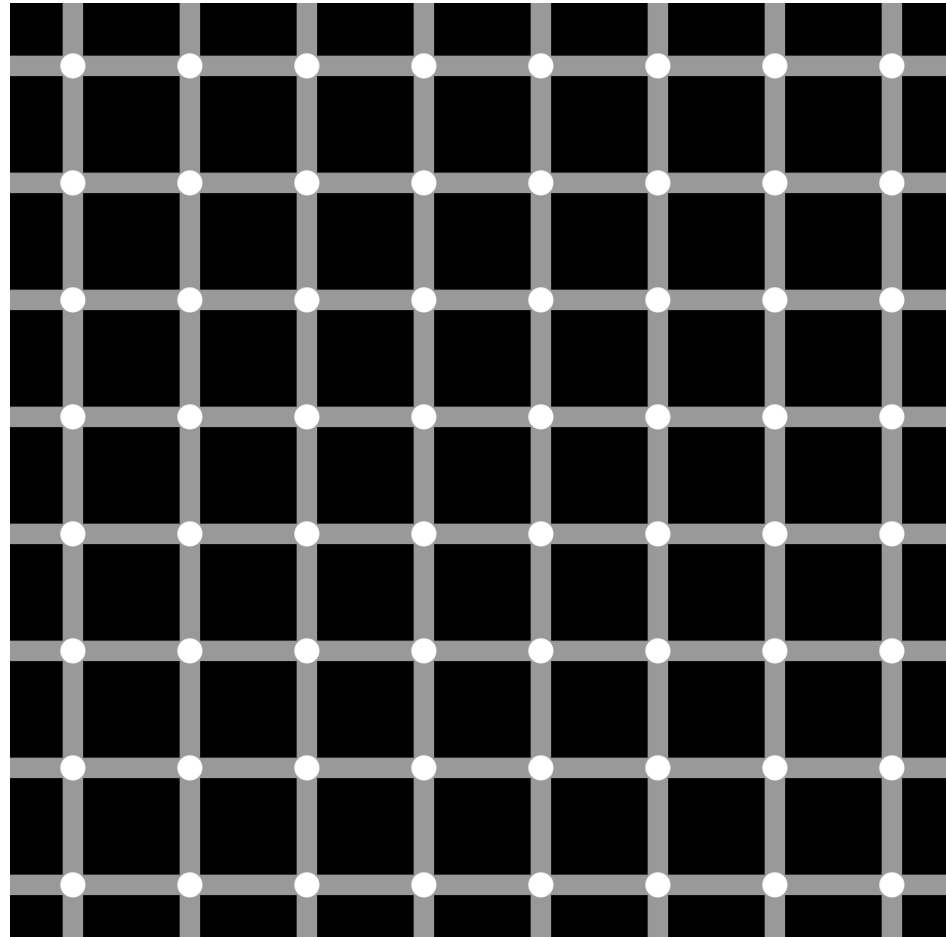
Perceptual limitations:

- **Troxler effect** occurs when one fixates on a particular point (red dot in figure right) for even a short period of time, an unchanging stimulus away from the fixation point will fade away and disappear.
- It is part of the general principle in sensory systems that unvarying stimuli soon disappear from our awareness.





Hermann Grid

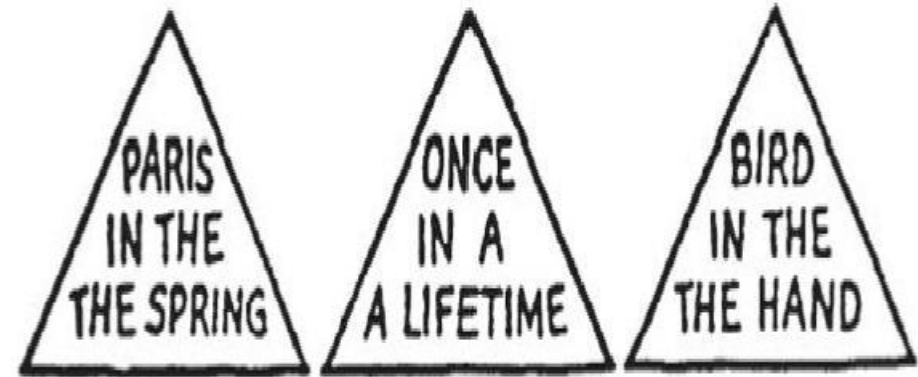


Scintillating Grid

Sensation, Perception and Attention are limited

Attentional limitations:

- We have finite attentional resources (can only focus on a limited number of things)
- Divided attention reduces performance on individual tasks and sustained attention degrades over time (vigilance decrement)
- Failures of visual awareness such as change blindness, repetition blindness, banner blindness, attentional blink, etc.



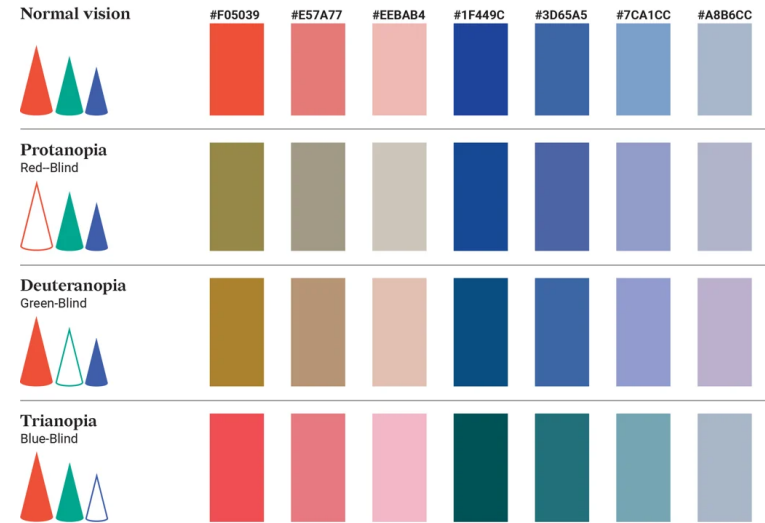
Movie Perception Test

from Levin & Simons (1997)

Effects on interface use

Sensory Limitations:

- Color blindness affects approximately 8% of men, requiring interfaces to not rely solely on color to convey information
- Small text or icons may be unreadable for users with visual impairments
- Audio cues may be missed by those with hearing limitations
- Touch interfaces may be challenging for users with reduced tactile sensitivity

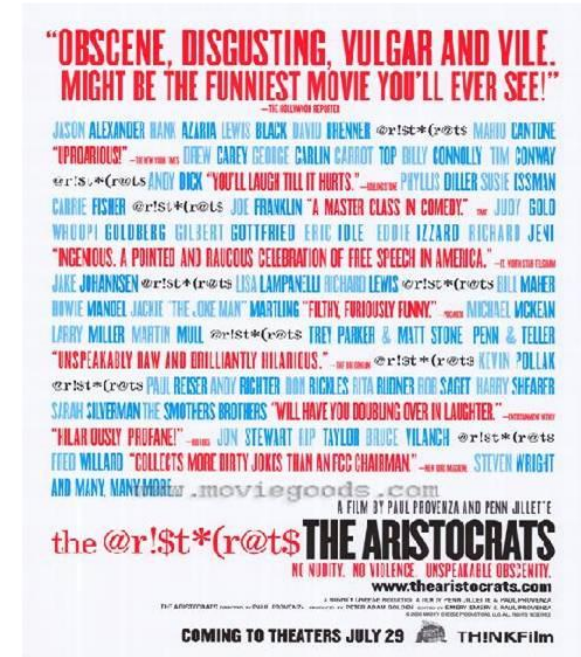


The first rule of making a palette for colorblind – avoid combining red and green. The picture above shows that for all of the types of color blindness there is a pair of colors that can be distinguished – red and blue – these are color blind safe colors.

Effects on interface use

Perceptual Limitations:

- Change blindness means users often miss interface updates that aren't explicitly highlighted (like new content appearing without animation)
- Banner blindness causes users to ignore content in locations typically used for advertisements
- Users struggle to find information when it doesn't match their mental models or expectations
- Gestalt principles affect how users group and perceive interface elements



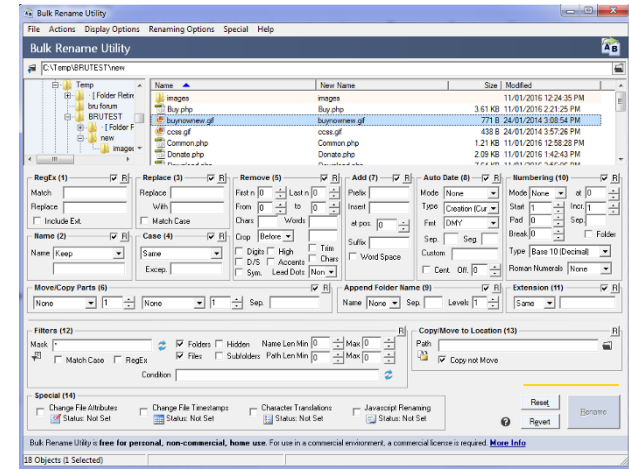
Bad use of Gestalt Principles

There is an attempt to separate the words using bold, but due to proximity, users will tend to read it as a whole sentence

Effects on interface use

Attentional Limitations:

- Notifications and alerts compete for limited attentional resources
- Multitasking between different interface elements reduces performance on each task
- Complex interfaces with many elements create cognitive load that exceeds working memory capacity
- Focus deteriorates during extended use sessions (leading to errors in critical systems)





Universitat
de les Illes Balears

2

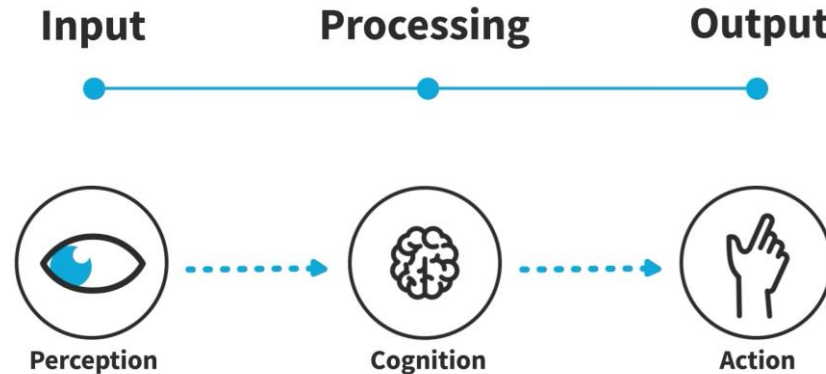
Information processing and human limitations

Human information processing

- Information processing refers specifically to the sequence of operations through which information flows in the mind — how we take in environmental stimuli, transform them into mental representations, manipulate these representations, store them, retrieve them, and use them to guide behavior. **It's a theoretical framework or model for understanding how cognition works.**
- **Cognition is the broader concept** — it encompasses all mental activities involved in acquiring, processing, storing, and using knowledge. This includes perception, attention, memory, language, problem-solving, reasoning, decision-making, and more. Cognition is essentially the entire landscape of mental functioning.

Human information processing

- The information processing theory likens the human brain's processing system to a computer. The computer (our brain) receives the data (sensory information), then processes it, either using it right away to perform a task or storing it in its files for later use.
- This isn't a perfect analogy for how the brain works. The human brain isn't simply a passive information processor.



Central components of information processing

Information processing theory has since been expanded and developed into several models. Each model is different and has its own strengths, but they all contain three central components:

- **Input.** This is the information that our brains receive from environmental stimuli. It can come through our senses (sight, smell, taste, touch, and hearing) or our memory (things we already know or have experienced).
- **Processing.** This is what happens to the sensory information once it's in our brain. We may take different actions with the knowledge, depending on what we want to do with it. For example, we may encode it to store in our long-term memory or keep it in our working memory for more immediate use.
- **Output.** This is what happens when we retrieve the information. We may recall it from our memory system or put it into action (e.g., through critical thinking or taking action).

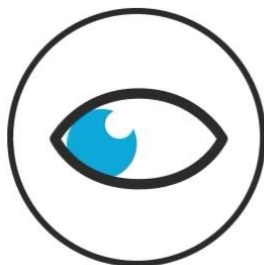
Computational Analogy



Input

Processing

Output



Perception



Cognition

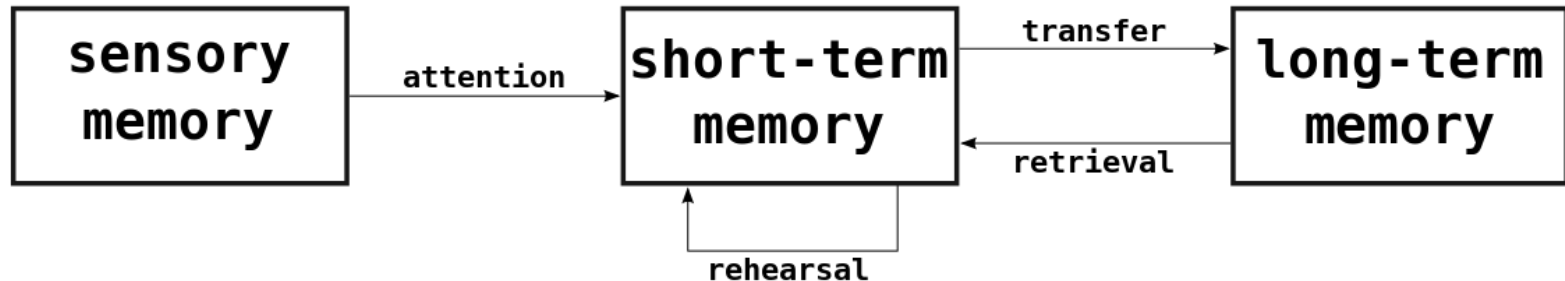


Action

Human information processing

The information processing theory **has been expanded into several models**. The Atkinson-Shiffrin model (multi-store model) is among the most noteworthy. The Atkinson-Shiffrin model was proposed by Richard C. Atkinson and Richard Shiffrin in 1968. The model suggests that information is stored in three separate memory stores.

MULTI-STORE MODEL



Human information processing

The Atkinson-Shiffrin model suggests that information is stored in three separate memory stores:

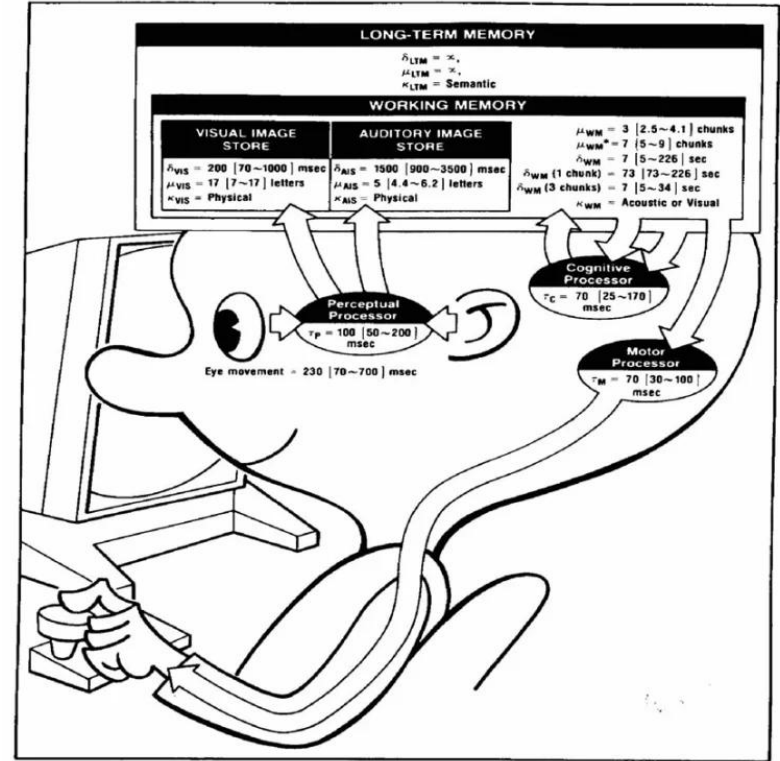
- **Sensory memory.** Information from external stimuli is first stored in the sensory memory, or sensory registers. This brief storage system holds information for a very short period of time, typically less than one second. From here, information either moves into short-term memory or is forgotten.
- **Short-term memory.** Up to 30 seconds of information can be stored in short-term memory. This information is then either released or transferred into long-term memory.
- **Long-term memory.** Long-term memory is a permanent storage system that can hold information indefinitely. Information in long-term memory is retrieved consciously or unconsciously.

Types of Memory



Human information processing – The Human Processor Model

The **Human Processor Model** was developed by Card, Moran and Newell in 1983. Card, Moran and Newell wanted to draw an analogy between the way humans perceive (process) and remember (store) things and the processing and storage of a computer as they believed that model of how human's process information would allow designers to predict what types of interface features would work.



I. S. MacKenzie, Human-computer interaction: an empirical research perspective, Second edition. Morgan Kaufmann, 2024.

Human information processing – Fitts's Law

- Fitts's Law is a predictive model of human movement primarily used in HCI that states that the time required to rapidly move to a target area is a function of the distance to the target and the size of the target.
- Fitts's law is used to model the act of pointing, either by physically touching an object with a hand or finger, or virtually, by pointing to an object on a computer monitor using a pointing device.
- Fitts's law has been shown to apply under a variety of conditions; with many different limbs (hands, feet, the lower lip, head-mounted sights), physical environments (including underwater), and user populations (young, old, special educational needs, and drugged participants).

Fitts's Law *demonstration*



An interactive presentation of Fitts's Law. Written in 2005 by [Marcin Wichary](#)
for the Vrije Universiteit.

click whenever
you're ready



← Previous

1. Title page



Next →

Human performance - Reaction Time

One of the most primitive manifestations of human performance is **simple reaction time**, defined as the delay between the occurrence of a single fixed stimulus and the initiation of a response assigned to it. An example is pressing a button in response to the onset of a stimulus light.

Operation	Time (ms)	
	Min	Max
Sensory reception	1	38
Neural transmission to brain	2	100
Cognitive processing	70	300
Neural transmission to muscle	10	20
Muscle latency and activation	30	70
Total:	113	528



Reaction Time Test

When the red box turns green, click as quickly as you can.
Click anywhere to start.

Human performance - Reaction Time

How to Interpret Your Reaction Time Test Results?

Decoding your Reaction Time Test results provides valuable insights into your neurological efficiency and sensorimotor performance. The following table breaks down reaction time ranges and their implications for cognitive processing and physical responsiveness:

Reaction Time (ms)	Performance Evaluation
150-200	Elite-level responsiveness, characteristic of top-tier athletes and individuals with extensively trained reflexes.
201-250	Excellent reaction time, indicating superior neural processing and motor control. Common in competitive sports and high-stakes professions.
251-300	Standard response range for adults, reflecting a well-functioning nervous system and typical cognitive processing speed.
301-350	Moderately slower reactions, often seen in individuals with less reflex-intensive lifestyles or those in older age groups.
351-400	Below-average response time, potentially indicating reduced cognitive processing speed or delayed motor reactions.
Over 400	Significantly delayed reaction time. If persistent, consider consulting a healthcare professional to evaluate potential underlying factors.

Human performance - Reaction Time

Weekly Assignment 2: The Python script [simple-detection-visual-pygame.py](#) in [chapter 12](#) of [Programming for Cognitive and Brain Sciences](#) is a simple detection experiment programmed with [Pygame](#). The task is simple: the participant must press a key as quickly as possible when a cross appears at the center of the screen. Download the script and run 4 tests for each member of your team.

The results of a test are saved in `reaction_times.csv` where one column is the delay presenting the cross and the other column is the reaction time both in milliseconds. Make [scatter plots](#) (delay vs reaction time) of the four tests of each team member ([all 4 tests in one figure](#)). Compute the [Pearson correlation coefficient](#) between delay and reaction time for all the tests.

Write a paper explaining your findings following the course guidelines.

Human performance - Memory Limitations

- **Capacity constraints:** Most people can only hold about 4-7 items in working memory at once (Miller's "magical number seven plus or minus two")
- **Duration limits:** Without active rehearsal, information in working memory fades within seconds
- **Encoding failures:** Not all information gets properly stored in long-term memory
- **Reconstruction errors:** Memories are not exact recordings but reconstructions influenced by schemas, beliefs, and subsequent information
- **Forgetting:** Information decays over time without retrieval practice

Human performance - Miller's Law

- The origin of Miller's law is in the paper ["The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information"](#) written in 1956 by cognitive psychologist George A. Miller.
- In the paper Miller discusses limitations in memory span, the longest list of **chunks** (the largest meaningful unit in the presented material that the person recognizes) that a person can repeat back in the correct order on 50% of trials immediately after the presentation.
- Miller observed that the memory span of young adults is approximately 7 chunks, it is often interpreted to argue that the number of objects an average human can hold in short-term memory is 7 ± 2 .

Recommended Reading: Herradura, H. ["UX myth, Miller's law: the magical seven was only an allegory"](#)

Memorize

1100110110011101

Memorize

1100 1101 1001 1101



Visual Memory Test

Memorize the squares.

Start



Universitat
de les Illes Balears

3

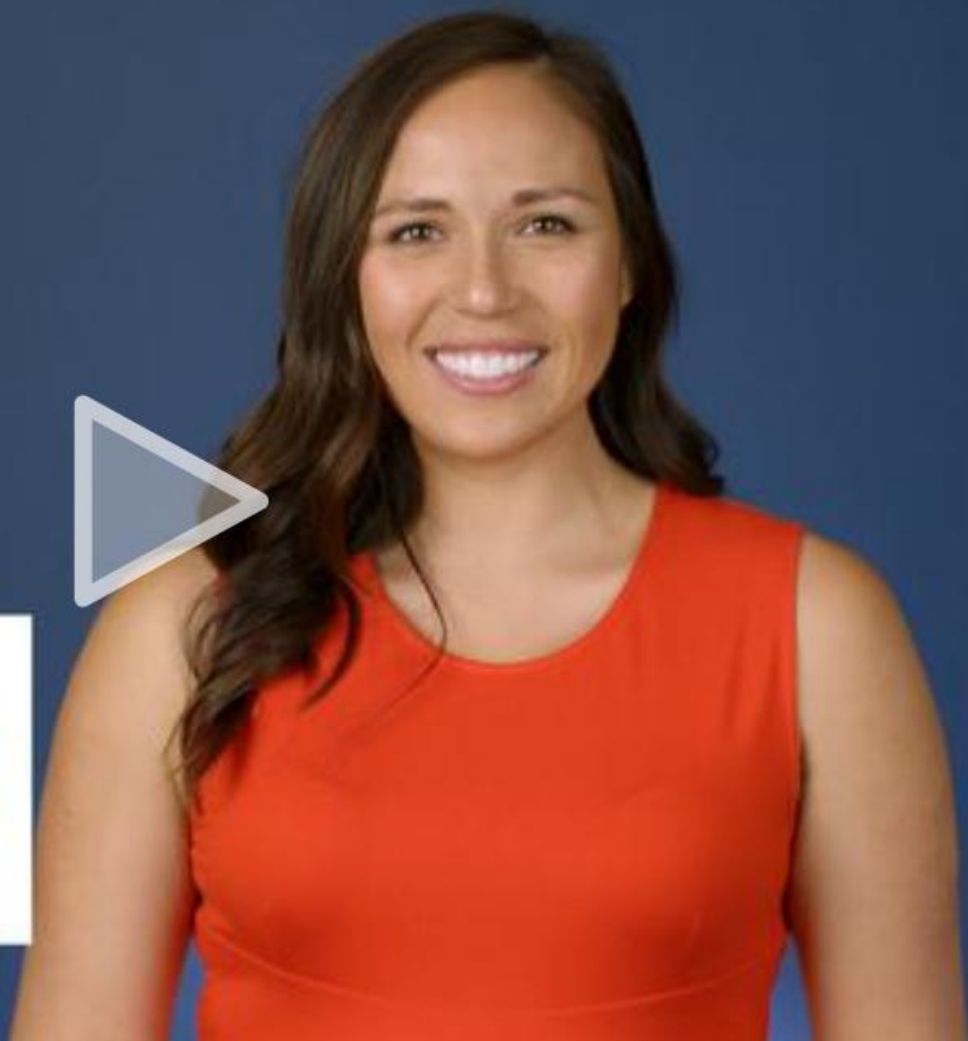
Mental models and their influence on design

Mental Models

- Mental models are representations of the world that help us understand complex concepts and make better decisions.
- They provide a framework for thinking and problem-solving, allow us to view problems from different angles and generate creative solutions, and help us become more effective thinkers and problem solvers.
- We create mental models based on past experiences, beliefs, and assumptions to understand how the world works.
- Mental models can be conscious or unconscious, varying in accuracy and usefulness depending on the context.
- Mental models are essential for decision-making, problem-solving, and learning, as well as effective communication and collaboration in group settings.

NN/g

Mental Models



Mental Models



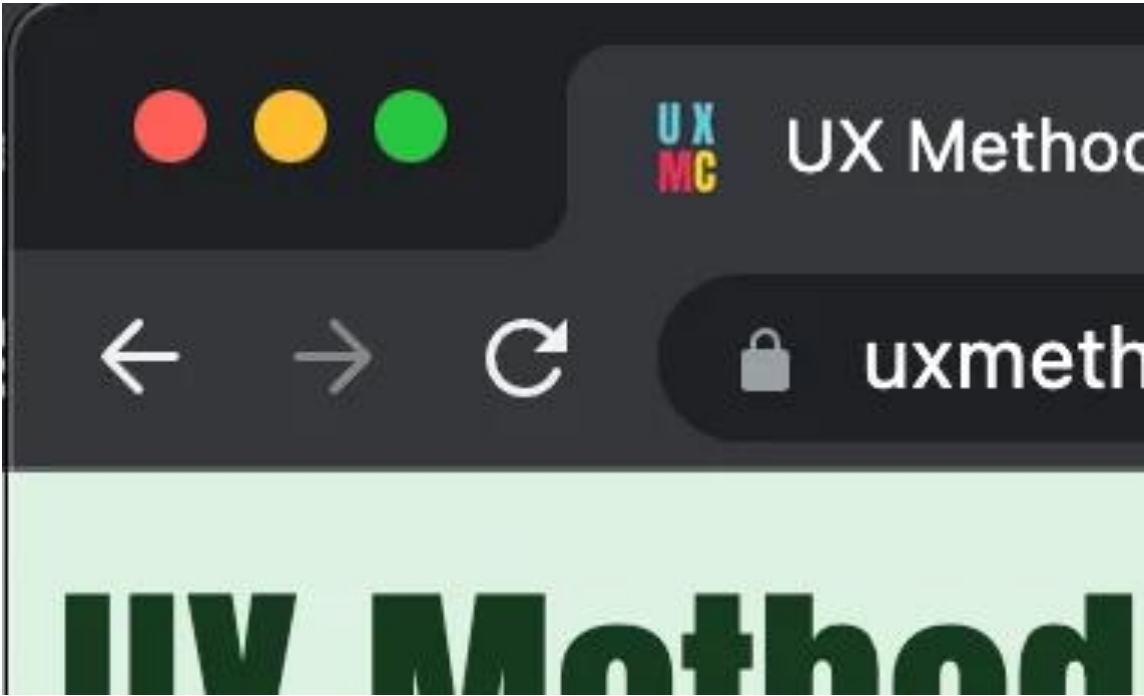
Norman Door

D. A. Norman, The design of everyday things, Rev. and Expanded edition. Basic Books, 2013.



Hamburger Menu

Mental Models



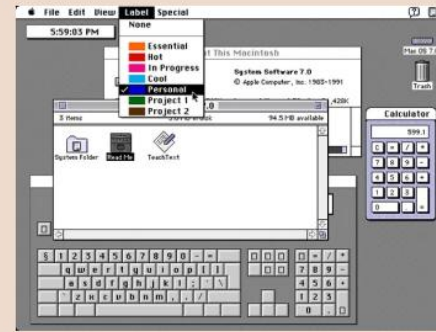
The Back and Forward Buttons



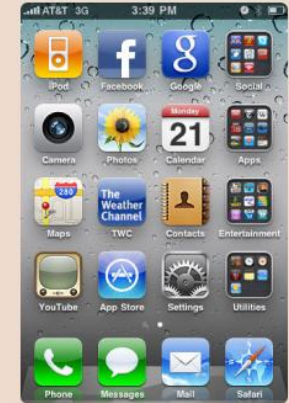
Ecommerce Shopping Carts

Mental Models - Skeuomorphic Design

- One way to leverage mental models is through skeuomorphic design, which involves incorporating real-world elements into digital interfaces.
- By imitating familiar physical objects, such as a bookshelf for organizing digital books, skeuomorphic design leverages users' pre-existing mental models.



Early GUI - Flat Style



Early skeuomorphic design

Mental Models - Skeuomorphic Design

Over time, the explosion of content richness highlighted the limitations of skeuomorphic design, leading to a shift towards flat design again, which prioritizes simplicity, information clarity, and efficiency.



Skeuomorphism



Flat design

Mental Models – The Desktop Metaphor

This foundational mental model transformed computing by mapping digital operations to familiar physical office objects.

- **Example in action:** When you "drag" a file into a "folder" on your computer, you're using a mental model based on physical document organization. The desktop, files, folders, and trash can are all representations that leverage our understanding of physical office environments.
- **Design implications:** This model made computers accessible to non-technical users by providing familiar reference points. Modern interfaces still maintain these metaphors even though many users have never used physical filing cabinets, showing how persistent mental models can become.

Mental Models – The Input-Output Model

This represents users' expectations about cause and effect in interfaces.

- **Example in action:** When a user clicks a button, they expect immediate visual feedback confirming their action was registered, such as the button changing color momentarily or an animation showing progress.
- **Design implications:** Interfaces must provide appropriate feedback for all user actions. Loading spinners, progress bars, and visual state changes are crucial for helping users understand system status and confirm their actions had the intended effect.

Mental Models – The Permission Model

This represents how users understand access and control in digital systems.

- **Example in action:** Users have developed mental models around permission dialogs, understanding that when an app asks to access their camera or location, they are granting specific capabilities to that application.
- **Design implications:** Permission requests should clearly communicate what is being accessed and why. Users feel violated when applications access more than what their mental model anticipated based on permission requests.

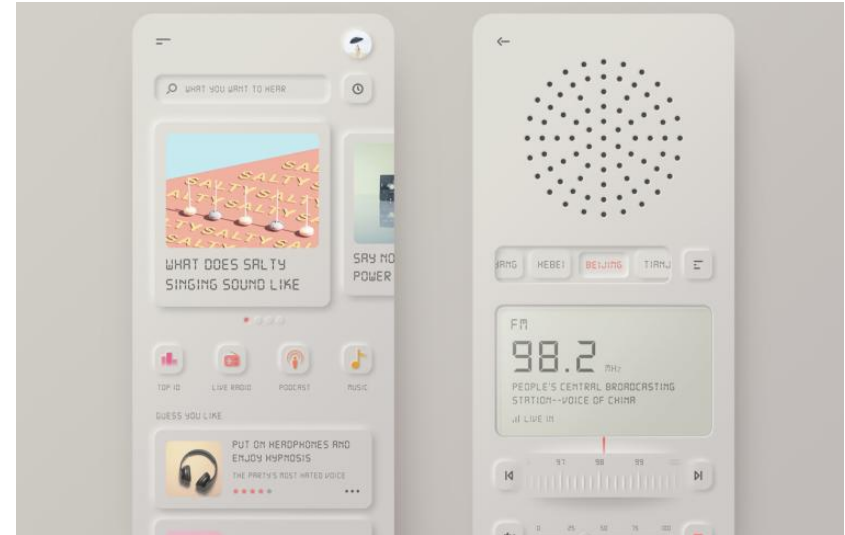
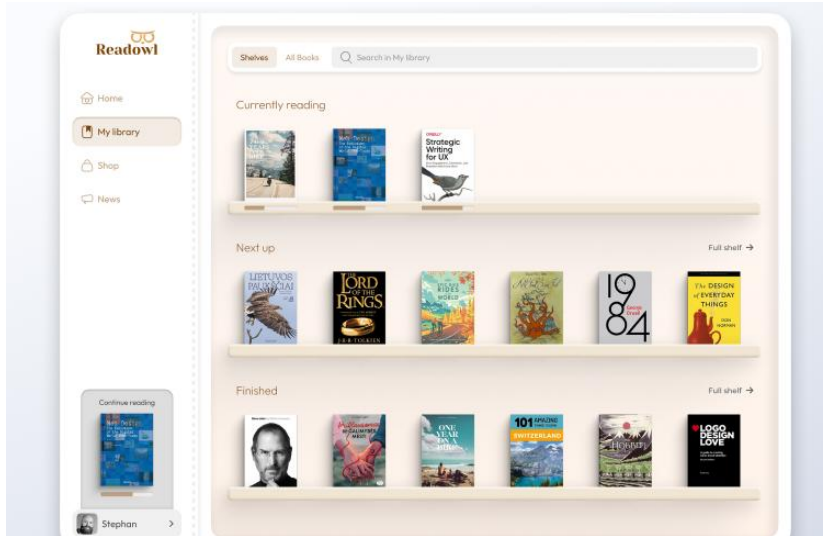
Mental Models – The Navigational Model

This represents how users conceptualize moving through digital information spaces.

- **Example in action:** Breadcrumb navigation in websites (Home > Products > Electronics > Smartphones) mimics the mental model of physical navigation through increasingly specific locations. Similarly, the back button in browsers reinforces a mental model of moving backward through a linear path.
- **Design implications:** Navigation should provide clear indicators of location and movement options. Users become disoriented when they can't determine where they are in an information hierarchy or how to return to previous states.

Mental Models - Skeuomorphic Design

However, in recent years we have witnessed a resurgence in skeuomorphic aesthetics.



The Role of Mental Models in Shaping Human-Centered Design

- Mental models serve as cognitive frameworks that guide user interactions with technology.
- These models profoundly influence design efficacy, usability, and behavioral outcomes.
- By understanding how users conceptualize systems and interactions, designers can create products that feel intuitive, reduce learning curves, and provide satisfying experiences.
- The most successful designs either leverage existing mental models or thoughtfully guide users in developing new ones that enhance their ability to interact with novel systems.
- The goal isn't necessarily to simplify everything to match existing mental models, but rather to create appropriate bridges between what users already understand and what they need to learn.

Recommended Readings:

- [1] A. Dix, J. Finlay, G. D. Abowd, and R. Beale, Human-computer interaction, 3rd edition. Pearson Education Limited, 2004. (Chapter 1)
- [2] I. S. MacKenzie, Human-computer interaction: an empirical research perspective, Second edition. Morgan Kaufmann, 2024. (Chapter 2)
- [3] D. A. Norman, The design of everyday things, Rev. and Expanded edition. Basic Books, 2013. (Chapters 2 and 3)



Universitat
de les Illes Balears